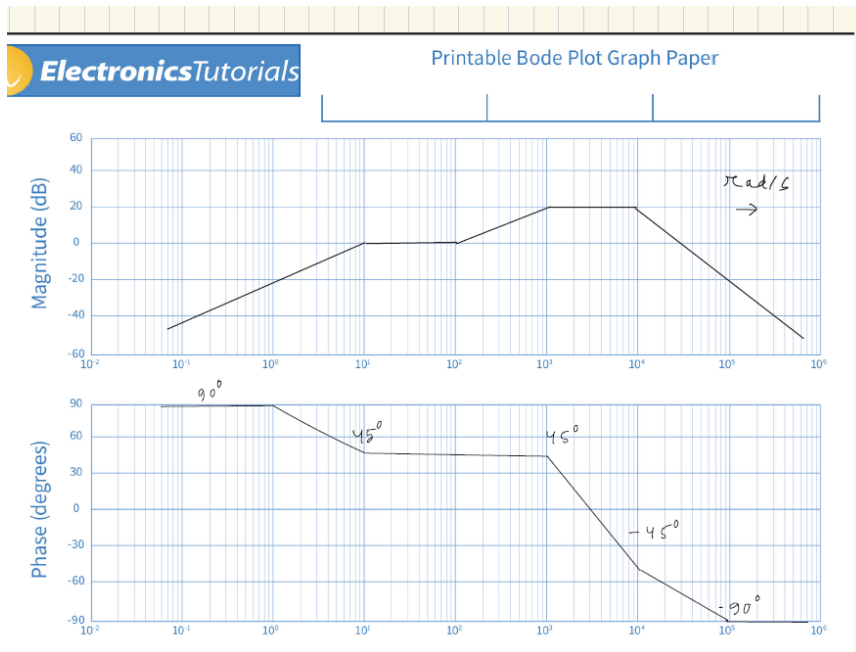
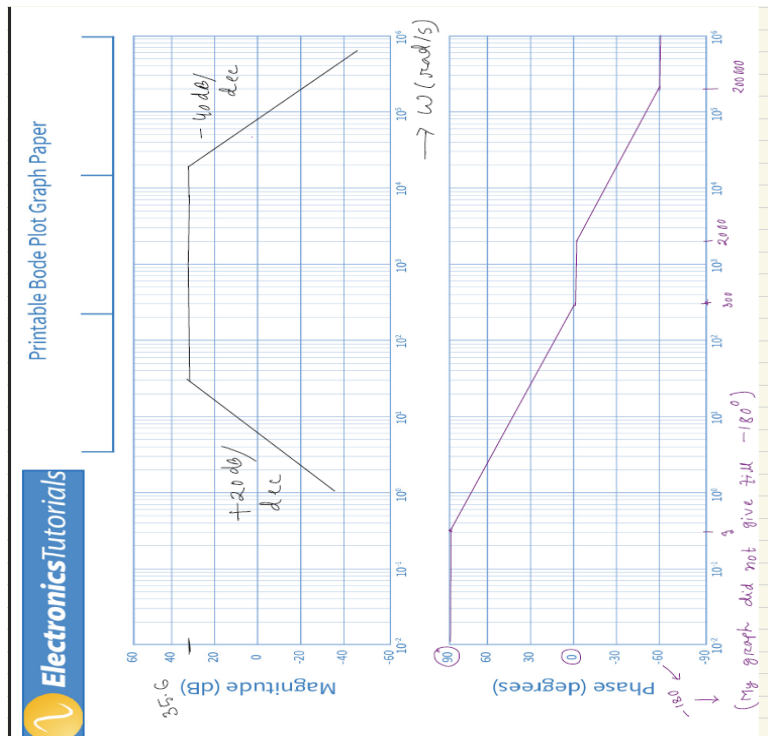


HW 3 ECE 3355

Problem 1:



Problem 2:



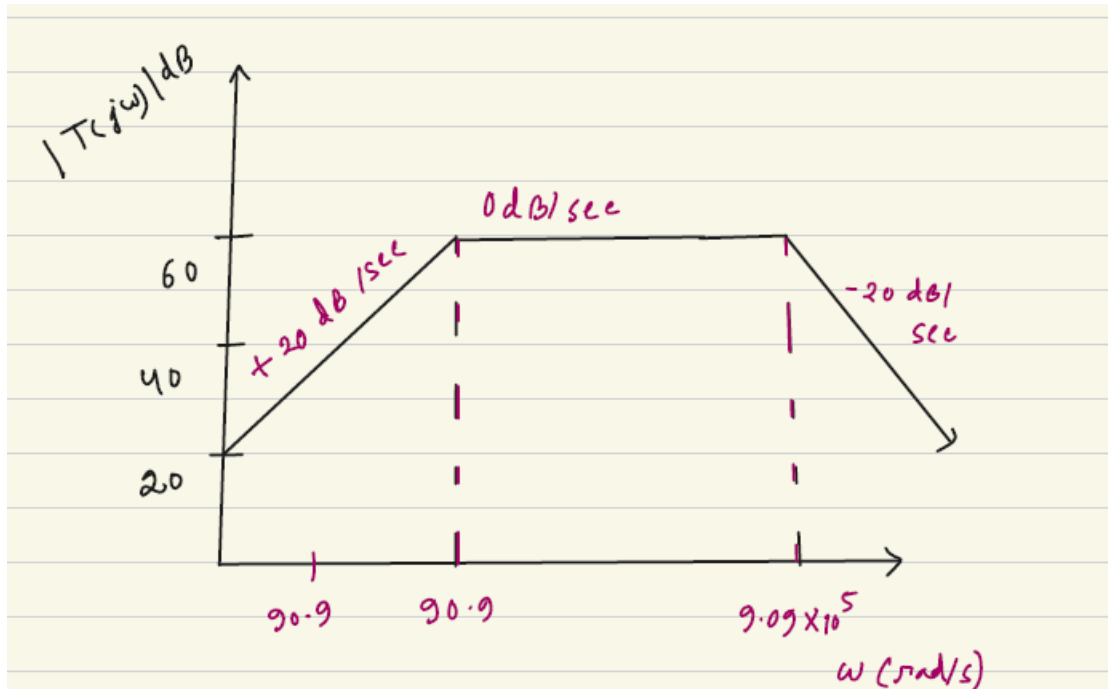
Problem 4:

$$T_i(s) = \frac{1}{s(100 \times 10^3)(10 \times 10^{-12}) + 1} \cdot \frac{1}{\left(\frac{s}{1000000}\right) + 1}$$

$$T_o(s) = \frac{-s10}{s0.011 + 1}$$

Cutoff frequency = 14.47 Hz

$$T_{(j\omega)} = \frac{-j\omega 10}{\left(1 + \frac{j\omega}{1000000}\right)\left(1 + \frac{j\omega}{90.9}\right)}$$

**Problem 5:**

a)

$$\frac{V_{out}}{V_{in}} = \frac{\frac{j\omega}{1000}}{\left(1 + \frac{j\omega}{1000}\right)\left(1 + \frac{j\omega}{10}\right)}$$

b)

$$\frac{V_{out}}{V_{in}} = \frac{\frac{jw}{1000}}{1 + jw0.2 + jw10^{-3} + j^2w^210^{-4}}$$

c)

$$\frac{V_{out}}{V_{in}} = \frac{\frac{3jw}{5000000000}}{\frac{jw}{10^9} + \frac{1}{10^6} + \frac{1}{10^6}} \cdot 61$$

Problem 6:

$$R_1 = 9[M\Omega]$$

$$C_1 = 3.33[pF]$$

Problem 7:

b)

$$A = 10 - \frac{100[v]}{0.3[v]\sin 60(rad / s)t + 10.5[v]}$$

c)

$$i_{out} = \frac{10(0.3[v]\sin 60(rad / s)t + 10.5[v])}{100\Omega}$$

d)

$$V_s = 0.5[v]$$